

Phytoremediation of North American Harmful Algal Blooms and Associated Cyanotoxins
Under Regional and Long-Term Feasibility Constraints: a Literature Review

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1. Abstract

Harmful algal blooms (HABs) are an environmental phenomenon increasing in prevalence, which ultimately poses an incredibly urgent environmental and public health concern. Exacerbated by eutrophication and anthropogenic climate change, these trends are only expected to worsen. Conventional HAB treatment methods, such as chemical algacides and mechanical removal, may reduce bloom biomass in short-term scenarios, they are often labor-intensive or carry ecological risks. This literature review assesses phytoremediation using 3 macrophytes (*Eichhornia crassipes*, *Lemna trisulca*, and *Myriophyllum aquaticum*) as an alternative for HAB and cyanotoxin management. Qualitative assessments evaluated the tendency to exhibit adverse effects for all respective macrophytes, and the literature was systematically reviewed to determine raw remediative efficiencies. *E. crassipes* demonstrates strong nutrient uptake and remediation potential; however, rapid proliferation leading to mat formation and regional restrictions limit application. *M. aquaticum* is a promising allelopathic agent capable of removing microcystins from the natural environment, but similarly to *E. crassipes*, its invasive behavior and fragmentation risk make it an unideal candidate. *L. trisulca* appears to maintain the strongest raw remediative potential and long-term feasibility due to its lower adverse-effect tendency, lack of restriction, and demonstrable ability for toxin removal and degradation.

2. Introduction

Harmful algal blooms (HABs) are an increasingly prevalent environmental phenomenon characterized by a proliferation of photoautotrophs, consisting of cyanobacteria, diatoms, and/or dinoflagellates. HABs, thriving most efficiently within slow or non-moving, warmer, nutrient-laden bodies of water, have been reported throughout the world and in every U.S. state (NOAA, n.d.; Pick, 2016), and climate change is only continuing to exacerbate incidence as conditions become more favorable (EPA, 2026).

Eutrophication, characterized by an overabundance of nutrients like nitrogen and phosphorus (N/P) within the water column, is one of the foremost causes of HAB events and will generally occur as a result of agricultural runoff (Daniel *et al.* 1998). Furthermore, HABs have been shown in the literature to induce hypoxia, or a partial lack of oxygen within the water column. Primarily occurring following bloom events when algal cells die, sink, and are decomposed, this process, by extension, removes dissolved O₂ (DO) from the water (ASCE, 2024). According to the EPA's Dissolved Oxygen "Indicators" Information Sheet, DO levels ranging from 1-3 mg/L are too low to support the majority of aquatic life, whereas levels below 5 mg/L are considered metabolically stressful.

HABs are known to produce bioactive metabolites referred to as cyanotoxins, capable of impacting the nervous, endocrine, and reproductive systems, among others (Cameán *et al.* 2017; Metcalf *et al.* 2021). These toxins impact recreational, economic, and environmental industries (Tokodi & Backović,

2024). In 2018, Florida's red tides (a type of HAB caused by *Karenia brevis*) caused an estimated \$2.7 billion in economic losses, primarily due to decreased revenues in various coastline businesses, such as hotels, restaurants, and bars (Alvarez *et al.* 2024). Annually, HAB events are estimated to cost between 10-100 million dollars in the United States alone, with some singular events averaging at tens of millions of dollars (NOAA, 2025).

Cyanotoxin exposure is a major contributing factor for adverse health effects in both humans and animals. Paralytic shellfish poisoning (PSP), for example, occurs in humans when exposed to saxitoxins (STX), a neurotoxic metabolite produced by cyanotoxins, by way of eating contaminated shellfish. Outbreak events have been found to coincide with major bloom proliferations - such as in Washington and Oregon in 2024, with over 40 confirmed cases of PSP, some requiring hospitalization (NCCOS, 2025). PSP can cause paralysis, respiratory failure, and ultimately, death, with symptoms such as numbness beginning in as little as 30 minutes (Jong, 2008). Domoic acid is a toxin known, additionally, for inducing amnesic shellfish poisoning (Zabaglo *et al.* 2016). In the past century alone, reports of cyanotoxin poisonings involving both animals and humans have increased ten-fold (Wood, 2016). With anthropogenic climate change actively increasing HAB incidence (EPA, 2026), this trend is only expected to worsen. Animals, such as livestock and pets, are especially vulnerable to HAB events as death can occur in as little as minutes to hours following exposure to produced cyanotoxins (Wolfe, 2021).

When cyanobacterial cells die, whether as a result of self-induced hypoxia or natural bloom decay, cell lysis may occur. Cyanotoxins, which are contained intracellularly, will then be released into the surrounding medium, where they can have detrimental effects on the environment (EPA, 2021). Facciponte *et al.* (2019) suggested that these toxins can then become aerosolized and may accumulate in the human upper respiratory tract and central airway, finding no evidence that distance from the nearest waterbody or time of year influenced exposure, which was interpreted as being indicative of a larger, more chronic issue. Metcalf *et al.* (2021) also found that various toxins (MCs, BMAA, ANTX-a) have been previously recovered from filter materials used in environmental/personal air sampling equipment, suggesting that airborne toxins may be present in human-occupied areas. If cyanotoxins are inhaled by humans, this may result in a rapid uptake into the central nervous system as they can bypass the blood-brain barrier via the olfactory nerve (Lucchini *et al.* 2012).

Microcystins (MCs) are hepatotoxic cyanotoxins and potent liver tumor promoters, of which there are over 300 recognized congeners (Spoof & Catherine, 2016; Gu *et al.* 2022). β -Methylamino-L-alanine (BMAA) is a neurotoxin associated with extremely increased incidences of ALS/Parkinsonism-Dementia Complex in Guam (Plato *et al.* 2022). Anatoxin-A (ANTX-a) is also called Very Fast Death Factor (VFDF), which is a name given by Carmichael & Gorham (1978) following observation of cell death in mice in as little as 2-7 minutes post-intraperitoneal injection. Although there are many more cyanotoxins and thousands of possible congeners, the referenced toxins (BMAA,

MCs, ANTX-a) are well-studied and popular within the literature, though some gaps may exist.

Many conventional treatment methods to combat HABs have significant drawbacks. For instance, in 2014, Lake Isabella, CA experienced a fish kill following a chemical treatment effort targeted towards cyanobacteria using a copper(II) sulfate pentahydrate-containing algaecide. Nearly 190,000 bluegill were found deceased shortly after application (Meek, 2014). Physical methods (such as netting or skimming the water's surface) can be laborious and ultimately ineffective, as any algal cells left behind can then re-proliferate. Therefore, biological treatment methods, such as phytoremediation, often show the most promise in terms of HAB remediation potential.

Phytoremediation, defined as using plants and their associated microorganisms to capture, detoxify, and/or degrade environmental contaminants, is a cost-effective alternative to many traditional treatment methods. Often, adverse effects are less common than with other techniques, such as chemical methods (Zamel *et al.* 2022). The literature has shown that macrophytes (aquatic plants) can metabolize contaminants analogously to an animal's liver. This is the Green Liver concept, as was popularized by Sandermann (1994). It most generally applies to non-rooted plants. Rooted macrophytes will often co-occur alongside associated microbial communities manifesting as biofilms capable of further assisting with phytoremediation (Ijoma *et al.* 2024). Some ways this is done is by improving toxin/chemical resistance, encouraging plant growth, and/or enhancing nutrient uptake (Ma *et al.* 2015).

However, like all HAB treatments, phytoremediation is not a universally applicable solution. The effectiveness of phytoremediation solutions varies depending on a variety of factors, including but not limited to: the macrophyte's nutrient uptake capacity, toxin resistance, tendency towards invasive behavior, and mechanism(s) of action. A macrophyte may demonstrate strong inhibitory effects on cyanobacteria, yet remain impractical for implementation on a larger scale due to invasive behavior or difficult management requirements. For example, the common water hyacinth (*Eichhornia crassipes*) has been shown to, despite its remediative efficiency, have the capability of "blooming" and forming a surface mat, which may induce hypoxia and harm economic industries (Otieno *et al.* 2019).

3. Macrophyte Classification Framework

Ultimately, three (3) macrophytes were identified as prominent potential phytoremediative agents for HAB mitigation. These plants were chosen on the basis of having demonstrated measurable results, in some form, within the literature search following exclusions.

In the interest of facilitating cross-study comparison, in addition to quantitative data, each identified macrophyte (*Eichhornia crassipes*, *Lemna trisulca*, *Pistia stratiotes*, *Spirodela polyrhiza*, *Ceratophyllum demersum*, and *Myriophyllum aquaticum*) was qualitatively classified using a predefined trait framework (Table 1). Macrophytes were characterized based on the following

characteristics: growth form, being free-floating, submerged, or emergent, mechanism(s) of removal/suppression, legality/invasive status (and therefore regional availability), and tendency towards adverse effects.

Table 1: Macrophyte Trait Framework				
Scientific Name	Growth Form	Remediation Mechanism	Restricted in North America?*	Tendency to Exhibit Adverse Effects?
<i>Eichhornia crassipes</i>	Free-floating	Rhizofiltration	Yes, 12+ states	Extreme
<i>Lemna trisulca</i>	Submerged	Bioaccumulation	No	Low-Moderate
<i>Myriophyllum aquaticum</i>	Emergent & submerged	Allelopathy	Yes, 20+ states & Canada	Moderate-High
*Restrictions were assessed based on IWGS' Regulated and Prohibited Aquatic Plants List (2018); information compiled from both Federal and State regulations, either available online to all or directly provided by State entities.				

A macrophyte was classified as free-floating if, within the natural environment, the plant floats on the surface of the water column with no sedimentary anchor. Conversely, if the plant grows completely underwater, it is characterized as submerged. An emergent macrophyte is one that grows rooted in shallow water, with the majority of its shoot system extending beyond the column.

A rudimentary Tendency to Exhibit Adverse Effects examination was conducted for each of the three (3) macrophytes analyzed. Free-floating macrophytes reside on the water column surface and are therefore more likely to proliferate and form mats, analogous to the surface scums produced by HABs in terms of their environmental impacts. Submerged and emergent macrophytes also maintain the potential to outcompete other plant species. While submerged macrophytes may be rooted in sediment, thus keeping them stationary, free-floating macrophytes are often mobile and can rapidly cover large expanses of water, potentially making removal challenging (Sender & Róžańska-Boczula, 2025). Moreover, federal and state restrictions (if applicable) were considered in classifying each macrophyte, as they are generally both indicative of the potential for harm and the region(s) wherein they can be applied (IWGS, 2018).

A macrophyte exhibiting an extreme tendency is one with a strong likelihood of causing serious ecological problems if deployed at scale - before, during, and/or after application. A high tendency implies that the macrophyte is prone to multiple adverse-effect pathways, and therefore requires active management while in use, with the additional possibility of subsequent conclusionary harm without sufficient techniques (i.e., a difficulty in removal of the macrophyte itself from the water column). A phytoremediative agent was

characterized as moderate if there are possible non-prominent adverse-effect pathways (such as a moderate spread risk or moderate removal burden). Similarly, if a macrophyte was classified as low tendency, it maintains a low likelihood of disruption under feasible deployment ranges.

4. *Eichhornia crassipes*

The common water hyacinth (*E. crassipes*) is a free-floating macrophyte native to the Amazonian basin. Its primary mechanism of action is rhizofiltration (Table 1), or the taking up of contaminants into the plant's biomass via its root system. This macrophyte is a widely applicable phytoremediative agent, as it is capable of bioremediating wastewater contaminated with heavy metal, organic, and inorganic pollutants - including N/P (Ayanda et al. 2020). Additionally, this plant is a well-known hyperaccumulator, meaning it can safely store a significantly higher concentration of pollutants than many other plants, making it an ideal candidate for phytoremediation (Rascio & Navari-Izzo, 2011). However, under eutrophic conditions, the macrophyte itself grows at an exponential rate, with a doubling time (DT) of approximately ten days (Soeprubowati *et al.* 2021). Due to this, in synergy with HAB inhibition, the plants can grow to outcompete even the cyanobacteria present within the water, creating a hyacinth mat. Hyacinth mats are dense surface scums (which can be up to 2 meters thick) made up of many individual hyacinth plants, obstructing the water column and potentially inducing hypoxia (Gupta & Yadav, 2020). Water hyacinth seeds can remain viable in the natural environment for nearly 20 years, which calls attention to the issue of post-remediative plant removal and the potential for re proliferation (Zulkeflee et al. 2021). Therefore, environmentally safe phytoremediation by way of *E. crassipes* requires significant active management, including both active observation to mitigate the chances of matting and adequate removal techniques to avoid re proliferation. If not properly disposed of, water hyacinth wastes can pose a considerable threat to the environment as accumulated contaminants may be re-released during decomposition.

4.1. Morphology & Uptake Dynamics

E. crassipes grows an incredibly complex, filamentous root system. Capable of growing up to 2 feet long and found hanging from the plant's underside, it provides significant surface area to support the flourishing of microbial communities, which can aid in phytoremediation. When *E. crassipes* is grown in low-P conditions, it adapts accordingly to enhance nutrient uptake. Yu & Xe (2003) found that, under these conditions, the lateral root grew 2.4-fold longer and nearly 2 times as dense while decreasing its circumference, with a thinner and longer root system generally increasing nutrient uptake from the environment. Yu *et al.* (2004) showed over a period of 8 weeks that, when *E. crassipes* is exposed to an environment with increasing nutrient concentrations, plant growth is stimulated and nutrient uptake rates are optimized. As eutrophic lakes often have consistent point sources of nutrients (e.g. fertilizers and applied

animal manure on farms), phytoremediation, in practice, requires a plant with the ability to adapt to actively increasing N/P concentrations.

E. crassipes has been shown to induce caspase-3-independent cell death in *M. aeruginosa* as a result of oxidative stress caused directly by the macrophyte. This contrasts with previous findings in the literature and indicates *E. crassipes* exhibits a unique mechanism of *Microcystis* suppression; for instance, Ross *et al.* (2006) concluded that cyanobacterial caspase-3 activity may be responsible for induced cell death in *M. aeruginosa*. Cell lysis was extensively observed in the control group containing no macrophyte, showing a 5.5-fold increase in extracellular cyanotoxin concentrations. The remediation group, in comparison, showed an overall decrease, indicating minimal cell lysis in practice (Zhou *et al.* 2014). This, however, does not account for the possibility of cell lysis during a HAB's decline phase, wherein algal cells die and decompose.

Jayaweera & Kasturiarachchi (2004) showed that the water hyacinth's growth rate is optimal when the plants are approximately 5-6 weeks old, with nutrient uptake also being optimal during this period of increased growth. A phytoremediative period of 3 weeks was recommended to allow for remediation, noting 100% removal of both N/P via assimilation by the end of this period. Qin *et al.* (2017) demonstrated that it was possible to transfer 87% of the intracellular cyanobacterial N pool (specifically from *M. aeruginosa*, an exceedingly prevalent hepatotoxic HAB-causing strain) from algal cells into the water hyacinth plant over a 4 week period of co-cultivation. N was accumulated in the roots and redistributed to organs such as the stalks and leaves, likely to prevent oversaturation of the root hairs. Imaka & Teranishi (1988), in one of the first novel studies concerning water hyacinth growth dynamics, found that the plant's N/P uptake rates were correlated with overall environmental N/P content by the Michaelis-Menten-type equation. As the concentration of N/P available to the plant rises, nutrient uptake increases until reaching a boundary, which is ultimately determined by i.) temperature and ii.) plant density. It is likely that, should the 1988 experiment be replicated with consideration to Jayaweera & Kasturiarachchi (2004)'s findings, nutrient uptake will be further enhanced, and adverse effects may be mitigated, which would help to enable sustainable application of the water hyacinth in future bloom events.

4.2. Microbial Association

Using 16 S rRNA amplicon sequencing, a technique for identifying and comparing bacterial composition, Singh *et al.* (2024) evaluated the microbial diversity of collected *E. crassipes* root samples. *Proteobacteria* was found to make up 79% of all bacteria found in the samples, followed by *firmicutes* (8%) and cyanobacteria (8%). The presence of cyanobacterial cells suggests the potential for bioaccumulation of the algal biomass itself within the macrophyte. While this can decrease algal biomass in the short term, it may pose concerns for macrophyte disposal as improper management - and subsequent reintroduction into a new environment - could in turn form a new HAB. *Epsilonproteobacteria* was the most prevalent class observed at 39%, followed by *Betaproteobacteria* at 18% and *Gammaproteobacteria* at 15%. Certain members of the

S. no.	Alpha-diversity indices	<i>E. crassipes</i> root zone	Lower quartile	Upper quartile
1.	Shannon index	3.492	3.487	3.498
2.	Simpson index	0.9001	0.8994	0.9008
3.	Chao-1	719.5	696.3	742.7

Table 2: (Singh *et al.* 2024; Table 1) α -diversity indices of the entirety of *E. crassipes* root system, as well as lower/upper quartile of root zone. Ranges, respectively (1-3) are: 0-~3.5; 0-1; 0-30,000+. Higher values generally indicate a higher biodiversity.

Betaproteobacteria class (e.g. *Paucibacter toxinivorans*) are capable of degrading cyanotoxins like microcystins and nodularin, which are liver-damaging (hepatotoxic) metabolites (Rapala *et al.* 2005). This suggests that the water hyacinth may have the ability to degrade cyanotoxins via its intrinsically-associated microbiota, further highlighting its role as an effective phytoremediator. Singh *et al.* (2024) additionally evaluated *E. crassipes* root zone α -diversity values (using 3 indices: Shannon, Simpson, and Chao-1). These indices measure ecological diversity, relative abundance, and species richness, respectively. The Shannon index value of 3.492 and Simpson index value of 0.9 indicate extremely high biodiversity within the *E. crassipes* root zone. Chao-1 values of ~696 and ~742 in the lower and upper quartiles, respectively, suggest that the latter maintains a higher species richness than the former. Yadav *et al.* (2023) found (via functional metagenomics) that the water hyacinth root zone harbored over 140 pollutant-degrading enzymes, impacting pollutants such as hydrocarbons, plastics, and dyes, among others, suggesting that the macrophyte can be used in water laden with other pollutants as well as cyanotoxins.

4.3. Management & Adverse Effects

E. crassipes grows incredibly fast, which is the foremost reason it's regulated in 12+ U.S. states and listed as #32 of the top 100 within the Global Invasive Species Database (Invasive Species Specialist Group, 2026). Under eutrophic conditions and in favorable temperatures (~30° C), the macrophyte can reach a biomass density of approximately 10kg/m² within 50 days, contrasting significantly with its value under normal conditions at ~0.1kg/m² (Wilson *et al.* 2005). Under similar conditions, many cyanobacterial strains have DTs of anywhere from a few hours to 5-14 days (Wilson *et al.* 2006; Wendt, 2022). Despite their DTs, hyacinth plants will generally outcompete cyanobacterial strains (as well as other present photoautotrophs) due to shading, as they form

thick mats that prevent the sun from penetrating the water's surface. Hyacinth removal methods are often laborious, expensive, or ineffective. For instance, even biological controls (e.g. weevils) have extremely varying rates of success (Day *et al.* 2023). Therefore, the best way to approach remediation by way of *E. crassipes* would be to develop a repeatable, central approach to application that allows the plant sufficient time to uptake N/P without facilitating hyacinth matting.

5. *Lemna trisulca*

L. trisulca is a submerged, free-floating macrophyte colloquially known as the ivy-leaved, or star duckweed. Its primary mechanism of action (as pictured in Table 1) is bioaccumulation, or the accumulation of pollutants at a rate faster than the plant can metabolize them. It is a known phytoremediative agent capable of degrading and removing contaminants, including heavy metals and cyanotoxins (Ergönül *et al.* 2022). Duchnik *et al.* (2021) reported no morphological changes in the macrophyte following cyanobacterial exposure and that a long-term 35-day cocultivation period with cyanobacteria was feasible. In cultivation with *R. raciborskii*, the macrophyte both decreased overall CYN concentrations by ~35% and inhibited cyanobacterial growth by increasing the medium's N:P ratio, as it unevenly absorbs phosphate ions in comparison to nitrates, with the former being the limiting nutrient for cyanobacterial growth. *L. trisulca* is native to North America and is therefore not restricted for purchase or transport, contrasting with many other phytoremediators, making application easy. The macrophyte generally exhibits fewer adverse effects than other macrophytes like *E. crassipes* and *M. aquaticum* given its characteristic as a submerged macrophyte, which will generally form surface scums at a lesser frequency than free-floating or emergent plants. However, this is still possible, and thus removal methods such as surface skimming or netting may be required. When combined with wind-driven booms, which are used to concentrate the plants, this removal method can be incredibly effective, albeit time-consuming.

5.1. Morphology & Uptake Dynamics

When flowering and fruiting, *L. trisulca*'s leaves emerge from the surface of the water, but they remain submerged otherwise, often up to 12-14m below the surface (Sree *et al.* 2023). The macrophyte maintains a single, threadlike root, which may be absent altogether in some specimens (Minnesota Wildflowers, n.d.). Kaminski *et al.* (2014) showed the potential for removal and degradation of ANTX-a by *L. trisulca* with HPLC, finding that, in media containing 50 µg ANTX-a, 95% of the total toxin concentration was removed. In media containing 250 µg ANTX-a, 86% was removed. It was found that degradation of ANTX-a by way of *L. trisulca* was also possible, and that the plant can effectively metabolize the toxin, with intracellular content decreasing by 76% within 14d following a transfer to clean media. Toxin release to the medium was negligible (an increase from 3.5% to 5.1%) over this period, indicating low rates of cell lysis. *E. crassipes*, in comparison, is known to induce cell lysis in cyanobacteria, which may cause toxins to release into the medium (Zhou *et al.* 2014). Slight leaf degeneration, or

chlorosis, was also observed in *L. trisulca* itself; however, the negligible medium toxin release indicates a lack of toxin bioaccumulation within the leaves. A marginally lower chlorophyll content (decreased 9% compared to controls) following co-exposure with different ANTX-a concentrations was also observed, indicating a slight decrease in overall photosynthetic activity. Kaminski *et al.* (2023) reported that, while *L. trisulca* showed the ability to uptake ANTX-a from the medium (40-70% of the extracellular pool), it increased extracellular medium release of toxin CYN from 0.39 to 4.23-fold, compared to the control. Overall, it was reported that toxin release in the medium ranged from 0.94 to 28.86-fold throughout all experimentation depending on cyanobacterial culture makeup. This contrasts with the Kaminski *et al.* (2014) finding of negligible toxin release. Kaminski *et al.* (2013) suggested that increased temperatures may lead to enhanced ANTX-a degradation.

L. trisulca was found to have reduced both cyanobacterial biomass and chlorophyll-a (ch-a) content, which is used as an indicator of the former, in cocultivation with strains *D. flos-aquae*, *M. aeruginosa*, and *R. raciborskii* by 8, 12, and 3-fold (in comparison to the control), and by 17.5, 4.3, and 32.6-fold, respectively (Kaminski *et al.* 2021). During cocultivation, nitrate uptake was mostly observed during the first 24 hours, with the concentration remaining relatively constant thereafter. *L. trisulca* absorbed 16% of the total nitrate pool, with cyanobacterial strain *R. raciborskii* absorbing 69%. With the exception of *D. flos-aquae*, phosphate concentrations decreased less when cyanobacterial strains were cocultivated with *L. trisulca* than when in single cultivations. This suggests that, although *L. trisulca* does maintain a relatively high growth rate, it may not compete as effectively for phosphates as it does nitrates, allowing some cyanobacterial strains (which favor phosphates) to sustain partial growth. This may provide some explanation as to why *R. raciborskii* increased its biomass by 1.9-fold during the 14d cocultivation period, despite having decreased overall in comparison to the control group.

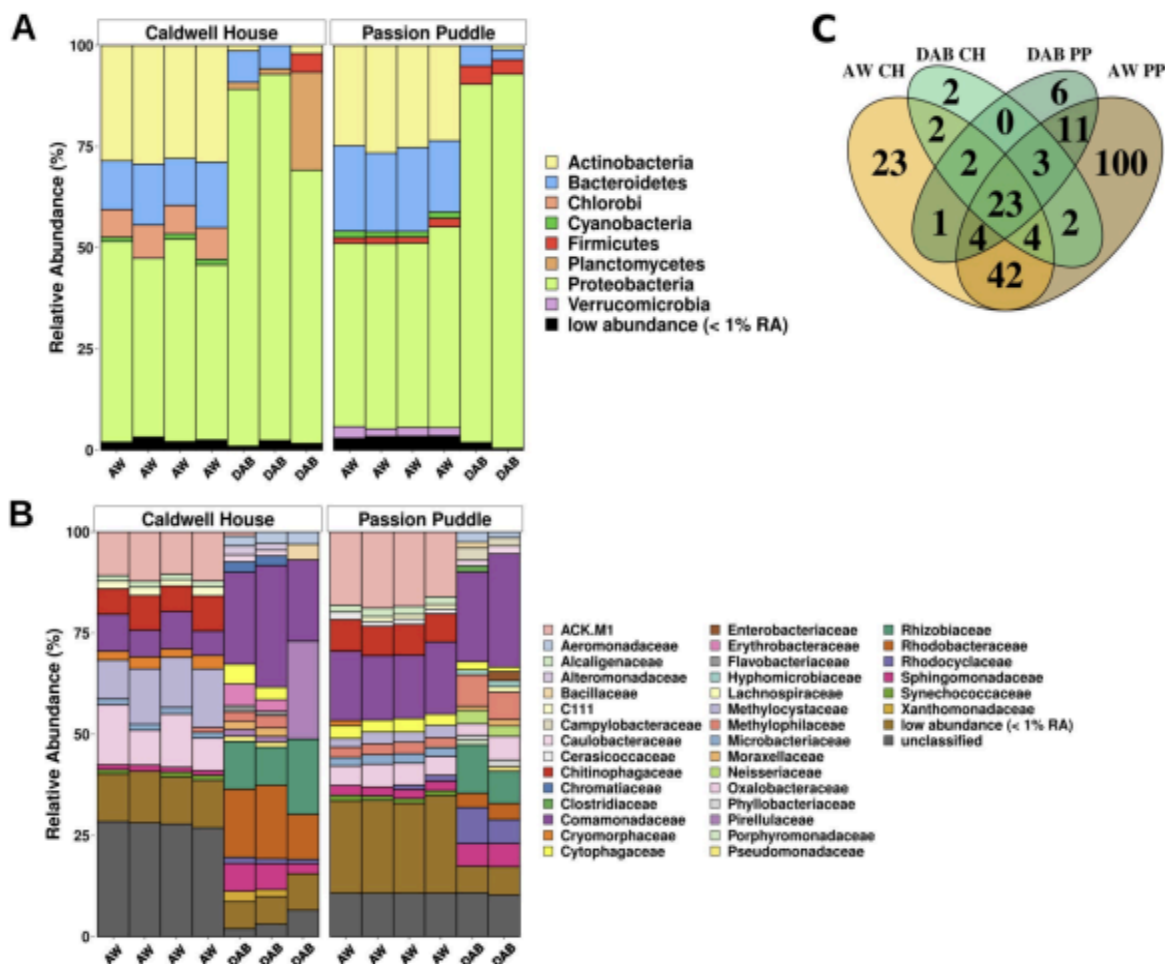
5.2. Microbial Association

Ike *et al.* (2017) presented findings that associated bacterial communities are capable of promoting duckweed growth. Lam *et al.* (2020) used ALDEx2 (Analysis Of Differential Abundance Taking Sample Variation), a type of differential abundance analysis, to evaluate bacterial composition associated with duckweed plants (Figure 1). *Proteobacteria* were found to be the most dominant class (~86% on average), which reflects the trend observed for *E. crassipes*, further highlighting the possibility of their association with phytoremediation. Several taxa (*Acinetobacter*, *Agrobacterium*, *Azospirillum*, *Burkholderia*, *Caulobacter*, *Methylibium*, *Pseudomonas*, and *Sphingomonas*) were identified as being constituents of the core duckweed microbiome, and are also known for their roles in maintaining plant health. *Comamonadaceae*, a family with established phytoremediative capabilities (Sun *et al.* 2024), was the most dominant observed at an average value of ~25% relative abundance. Popa *et al.* (2019) additionally evaluated the microbial diversity of naturally occurring *L.*

trisolca samples. Using Shannon and Simpson indices, biodiversity was determined to be relatively high, despite the lack of prominent roots, at values of 2.78 and 0.955, respectively. Compared to *E. crassipes*, with respective index scores of 3.492 and 0.9, this indicates that *L. trisolca* hosts a lower overall species count with higher diversity, and that the majority of its microbiota are likely contained external to the root system.

5.3. Management & Adverse Effects

L. trisolca is a predominantly submerged macrophyte, freefloating just beneath the surface of the water column. Adverse effects of this specific macrophyte have not yet been comprehensively examined within the literature, as it has largely been assumed they would manifest to a lesser severity than other plants. Some duckweed species are known to proliferate excessively and potentially overtake bodies of water (Borisjuk *et al.* 2023). However, *L. trisolca*, characteristically being the only submerged *Lemna* species, likely would not exhibit this phenomenon as strongly as *L. minor*, for instance, given that it does not reside on the surface of the water column. *L. trisolca* has additionally been shown to increase dissolved O₂ in water which may mitigate eutrophication-induced hypoxia (Sender & Róžańska-Boczula, 2024). Traditionally, methods like netting and skimming have been widely used for duckweed removal in place of chemical methods, which can have adverse effects. In recent years, low-cost autonomous skimmers have been explored in the literature. Salcedo *et al.* (2024) proposed a low-cost unmanned surface vehicle designed to passively collect duckweed using computer vision and autonomous navigation systems, which stands out as a potentially long-term management method.



6. *Myriophyllum aquaticum*

M. aquaticum, or Parrot's feather, is a macrophyte native to the Amazonian basin. Exhibiting heterophylly, this plant is widely considered to display both emergent and submerged qualities. The plant's stem can grow upwards of five feet long and may emerge from the water up to a foot, with both emergent and submerged leaves. The macrophyte has rhizofiltrative capabilities, but its predominant mechanism of action is allelopathy (Table 1). *M. aquaticum* has been shown to secrete allelochemicals (phenolic compounds) capable of suppressing algal growth and ensuring the macrophyte's competitive dominance (Wang et al. 2017; Xuexiu et al. 2008). *M. aquaticum* has established phytoremediative capabilities, specifically in terms of treating heavy metals, nutrients, and organic pollutants (Yao et al. 2021). *M. aquaticum* exhibits a DT of ~6-7 days (Teodorović et al. 2015), and is able to outcompete native photoautotrophs. According to the IWGS, it is restricted in 20+ U.S. states due to its invasive tendencies, which can make application in North American contexts difficult.

6.1. Morphology & Uptake Dynamics

M. aquaticum has a highly robust root system. It is a rhizomatous plant, meaning it develops rhizomes, or horizontally-growing underground stems that facilitate further root growth and store nutrients. Zhuang *et al.* (2023) used high-throughput sequencing analysis to evaluate the response of *M. aquaticum* to phosphorus stress brought about by both an overabundance of phosphorus and a lack thereof. Growth rate was assessed, showing a marginal, largely insignificant decrease under high levels of phosphorus. Significant reductions in both root length and overall root count, however, were observed under high-phosphorus conditions. Under low-phosphorous conditions, adverse effects observed in the macrophyte were worsened significantly. This contrasts strongly with other macrophytes, such as *E. crassipes*, for example, which exhibit notable root growth and an increase in overall growth rates under lower nutrient concentrations (Yu & Xe, 2003; Yu *et al.* 2004). Wu *et al.* (2022) reported that, with increasing ammonium concentrations (a nitrate essential for plant growth), root length rapidly decreased in accordance with a double exponential decay equation. This further reinforces Xu *et al.* (2023)'s findings and suggests that this trend in *M. aquaticum* is not unique to phosphates, but nitrates as well.

Kitamura *et al.* (2023) evaluated the allelopathic effects, in terms of produced phenolic compounds, of *M. aquaticum* on *M. aeruginosa* and its produced toxins depending on the season. MC-LR concentrations were quantified using HPLC, and it was found that the cyanotoxin was effectively removed, achieving an efficiency rate of up to 89%, with an average efficiency of $54.77\% \pm 18.87\%$. In another study, over a period of 7 days, *M. aquaticum* was shown to be capable of removing MC-LR from its surrounding medium, exhibiting a 6.6% removal rate at $1\mu\text{g MC-LR}_{\text{eq}}\text{L}^{-1}$ and a 19.3% removal rate at $10\mu\text{g MC-LR}_{\text{eq}}\text{L}^{-1}$ (Silva de Assis *et al.* 2024). The effect(s) of cyanotoxins on *M. aquaticum* has yet to be comprehensively explored within the literature.

6.2. Microbial Association

Li *et al.* (2020) evaluated the microbial communities present on biofilms attached to the stems and roots of the macrophyte. They determined that proteobacteria, bacteroidetes, and fusobacteria were the most dominant phyla, making up 73-90% of detected microbes. Proteobacteria were, overall, the most dominant phylum in all samples. Luo *et al.* (2023) also evaluated *M. aquaticum* microbial communities, finding similar results, with the addition of running α -diversity indices (Shannon and Chao-1). Values were reported to be 5.24 and 2,168, respectively, indicating extremely high diversity within associated microbial communities, more so than both *E. crassipes* and *L. trisulca*.

6.3. Management & Adverse Effects

M. aquaticum behaves similarly to *E. crassipes* in terms of its potential for adverse effects. Given its growth rate, the primary concern for management is the potential for plant fragmentation during removal and subsequent

reproliferation. Traditionally, treatment methods consist of i.) seining or raking the water, ii.) lowering water levels to expose the plants to extreme environmental conditions, or iii.) applying herbicides, although all referenced methods maintain their own respective drawbacks. In the case of seining or raking, fragmentation can cause reproliferation. Water body drawdown may not, in all cases, be enough to manage the population as plants can retain moisture. Herbicides, like algicides, can have detrimental effects on the environment and animals alike (Hadfield *et al.* 2024). Similar to *E. crassipes*, the best management practice for *M. aquaticum* is prevention; a standard should be developed which allows for an adequate remediation period without facilitating overgrowth.

7. Discussion

Much of the existing literature on HAB phytoremediation fails to critically evaluate the long-term feasibility of utilized plants. Studies often focus instead on raw remediative effectiveness while acknowledging the possibility of long-term difficulties with little further analysis. Therefore, the intention of this literature review was to evaluate the remediative effectiveness of various macrophytes commonly used for HAB control through the specific lens of long-term feasibility in North American contexts.

Many macrophytes demonstrate the potential for HAB suppression and/or cyanotoxin removal, but raw remediative efficiency alone is not a sufficient measure of practicality in real-world scenarios. Through the lens of application in North American contexts, species with lower tendencies to exhibit adverse effects, fewer regulations, and more manageable downsides are ultimately more feasible than other macrophytes with potentially more pronounced short-term uptakes.

A reliance on controlled laboratory conditions (mesocosm studies) can be found within the literature. While useful for isolating individual mechanisms, this does not fully account for all environmental conditions that contribute to HAB events (i.e. temperature, nutrient loading, species composition, sedimentary interactions, etc.). Macrophytes may behave differently in a real-world lake system with active nutrient loading occurring, and future studies should therefore take this into account. Varying contributory conditions may allow for further exploration of macrophyte-cyanobacterial interactions, including but not limited to microbiota activity, nutrient uptake, and allelopathic secretion.

8. Conclusion

HABs are increasingly prevalent in the natural environment as anthropogenic global warming worsens. Conventional treatment methods, such as physical and chemical approaches, may be labor-intensive or carry severe adverse effects. Phytoremediation is presented within the literature as a sufficient alternative to the aforementioned techniques as many macrophytes effectively

starve algal cells by reducing nutrient availability, while some plants may be capable of directly suppressing growth or toxin production.

HAB phytoremediation should be evaluated through a feasibility-centered framework, rather than one centered around raw efficiency. In reality, the ideal phytoremediative agent is not one that removes toxins or nutrients to the greatest extent, but rather one that can be deployed on large scales, managed, and ultimately removed without inducing adverse environmental effects. *L. trisulca*, accordingly, appears to hold the strongest long-term potential specifically for North American HAB remediation.

Of the three analyzed macrophytes, *E. crassipes* demonstrated the strongest raw remediation potential, which can be attributed to its rhizofiltrative uptake capabilities as well as its resistance to cyanotoxins. The plant's associated microbiota may also assist with phytoremediation as suggested in the literature. However, its overall feasibility lies in its ability to be applied on larger scales: the plant maintains several adverse effect pathways (e.g. surface matting) and therefore, may not be the most ideal candidate.

L. trisulca, conversely, seems to be the most promising candidate when feasibility of application is considered. While *E. crassipes* may have a stronger raw uptake potential, *L. trisulca* is native to parts of North America (and is therefore unrestricted), is less likely to exhibit matting due to its submerged growth form, and it shows the ability to both mitigate hypoxia and reduce cyanotoxin concentrations. Its effects appear species-specific, however, as the literature has reported varying rates of cell lysis dependent on cyanobacterial makeup.

M. aquaticum exhibits allelopathic suppression of cyanobacteria and the ability to remove toxins like MC-LR, but its invasive behaviors severely limit its feasible ranges of application. Fragmentation is one of the foremost concerns with this macrophyte; in scenarios where full biomass recovery is not guaranteed, re proliferation is likely. It is therefore recommended that *M. aquaticum* is used for controlled, experimental studies in the future rather than open-water application.

9. References

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